Review Classmates: Code Design Peer Review

Review by January 6, 11:59 PM PT

Reviews 3 left to complete

Graph to store street data and implementation of the BFS algorithm on this graph to generate driving directions

by Denisenko Alina
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Please enter the description of the classes you created to support your graph structure (i.e. all of the code in the roadgraph package). For each class include a few sentences about the purpose of the class and how it helps store the data required. After your description of your classes, provide a brief justification (1 paragraph, 4-6 sentences) for your overall design. You should use the template provided in this assignment's instructions.

Class: MapGraph
Modifications made to MapGraph (what and why):

I implemented all required methods
I added to this class member variable *Map<GeographicPoint, MapNode> nodes* to hold graph representation. Using this variable it is easy to find a MapNode in the graph by its geo location.
I implemented bfs method using 2 help private methods *bfsSearch* and *constructPath,* that make code more readable.

Class name: MapNode
Purpose and description of class:
A class to represent a Node in a graph which is an intersection at a unique geographic point. It holds geo location variable and all out-edges from this node.

Class name: MapEdge
Purpose and description of class:
A class to represent a directed edge between two Nodes of a graph, that are already in the graph. An edge is a street segment, that connect intersections. It contains information about start and end node for this edge and some useful information about road, such as length, name and type.

I used an adjacency list representation of a graph: I hold for each Node a list of all its out-edges. And for BFS algorithm I have a Map with GeographicPoint - MapNode pairs. It helps me to pretty quickly find by geo position a MapNode in the graph and all its neighbors. I also used java8 features in my implementation.

Is their description useful in helping you understand and review their code? Note: You must finish your review of their code before you can really answer this question. We recommend you read their description, then answer the prompts below, then come back to respond to this prompt.

- (●) 1 pt
  Yes
- ( ) 0 pts
  No

Help your peer! Complete the following prompt: "The description would have been more useful to me if..."

Is not it a bit redundant to store the location for a node both in the map as well as in the MapNode object?

Please upload a zip file containing all of the code in your mapgraph package.

for-review.zip (https://s3.amazonaws.com/coursera-uploads/peer-review/NwQ9txnoEeWDtQoum3sFeQ/dc5bdef023d4fa6d33a7c6e959347250/for-review.zip)

Do the objects make sense and do the data and methods in those objects go together?

- 2 pts
  Yes
- 1 pt
  Sort of, but there are some aspects don't seem to fit
- 0 pts
  No, not really

Have they developed an appropriate number of classes?

- 1 pt
  Yes
- 0 pts
  They did not develop enough. They should have created more.
- 0 pts
  They developed too many. They should have created fewer.

Are their classes used appropriately in the MapGraph class to store the necessary data?

- 1 pt
  Yes
- 0 pts
  No

Do they use appropriate data structures throughout their code to store their data for efficient retrieval (e.g. is it fast to access vertices and edges when performing the search)?

- 2 pts
  Yes
- 1 pt
  Sometimes. But some data structures could be improved.
- 0 pts
  No

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Are class interfaces clean (appropriate access and methods, private data (or data structures) are not exposed)?

- 2 pts
  Yes.
- 1 pt
  Sometimes, but they could be improved.
- 0 pts
  No (or almost never)

Are methods are short and structured appropriately? E.g. do they use helper methods where appropriate?

- 2 pts
  Yes
- 0 pts
  Sometimes
- 0 pts
  No (or almost never)

Is their code easy to read? Consider the following when answering this question: Is there no redundant code? Do they use meaningful variable names and appropriate indentation and spacing?

- 2 pts
  Yes
- 1 pt
  It's OK, but it could be better
- 0 pts
  No, it's very hard to read

Do they include appropriate comments? That is: Does each method have a header comment that clearly describes what the method does, including its input and output? Are in-line comments used where needed? (note: good code should not need many in-line comments).

Help Center

- 2 pts
  Yes, always (or almost always)
- 1 pt
  Sometimes, but it could be better

0 pts
No (or almost never)

Help your peer! Explain your reasoning for any of your responses above. Remember to keep your comments constructive and respectful.

Why is the member nodes in the MapGraph class declared final? I am new to functional programming constructs in Java, so could understand the code better if more comments were given at appropriate places (e.g., why use parallelStream() etc.). Is not it a bit redundant to store the location for a node both in the map as well as in the MapNode object? Is the MapNode class necessary (since the locations are anyway stored in the map), in other words, would not just an adjacency list declaration like Map<GeoPoint, ArrayList<MapEdges> > suffice?

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